Benoit L. Marteau

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LinkedIn | GitHub | Portfolio | Google Scholar

SUMMARY

PhD student at Georgia Tech with **30+ publications** (9 first/co-first) at ACL Findings, CVPR/ICLR/NeurIPS workshops, and IEEE journals. Research focuses on **Trustworthy Data and AI**, developing methods to assess data quality at scale, audit model reliability, and enable faithful multimodal reasoning for real-world deployment. **IEEE Best Paper Award**. GPA 4.0. Seeking **Applied Science internship**.

EDUCATION

Georgia Institute of Technology — *Atlanta, GA*
Ph.D., Electrical and Computer Engineering — *2022 to Expected 2026*

- Trustworthy AI, Multimodal Reasoning, Computer Vision, Foundation Models, World Models
- IEEE ICIR 2022 **Best Paper Award** | **NSF-Google-EMBS Award** (2025) | NSF Travel Award (2024)
- IEEE BHI 2025 Young Professional Co-Chair | STAR-AI Symposium 2026 Organizing Committee

M.S., Electrical and Computer Engineering (Double Diploma with Arts et Métiers) — *2019*

Arts et Métiers ParisTech — *France*
Diplôme d'Ingénieur (M.S. Engineering) — *2016 to 2020*

- Dassault UAV Challenge Team Lead (2017): **3rd place**; pioneered deep learning autonomous navigation

EXPERIENCE

Graduate Research Assistant and Lab Manager — *Jan 2022 to Present*
Bio-MIBLab, Georgia Institute of Technology (Advisor: Dr. May D. Wang) — *Atlanta, GA*

- **Faithful Multimodal Reasoning** (ACL 2026 Findings, co-first): Designed Tree-of-Evidence, an inference-time beam search for auditable predictions. **Retains 98%+ of full-model AUROC with 5 evidence units**; fidelity error **-56% vs. greedy**, **-58% vs. LIME**; outperforms 70B LLMs with 109M params. 6 tasks, 3 datasets, 2 domains.
- **Trustworthy Data Quality at Scale** (IEEE BHI 2024 and 2025, first author): Modernized multi-site pediatric warehouse to OMOP CDM v5.4; built Python quality tool on MS Fabric replacing R/Java stack. **1,100+ tests over 750M data points** (83.2% pass). Extended with METRIC framework.
- **Generative AI and Computer Vision** (ACM BCB 2023, co-first; NeurIPS DGM4H 2023; ISBI 2025, co-first): Diffusion vs. GAN for rare-disease pathology: **sFID 97.6 vs. 337.7** (3.5x). Synthetic H&E from 58-marker immunofluorescence via U-Net (**PCC 0.842**). Dataset cartography across internal and external cohorts.
- **Real-Time ML and Digital Twins** (IEEE ICIR 2022, co-first, **Best Paper**): Cloud digital twin with real-time sensor streaming (ESP32 to Azure IoT Hub to Function App ML inference); photogrammetry-based 1:1 environment reconstruction; deployed as web application.
- **Lab Management**: Manage cloud infrastructure and GPU compute for 50-person research lab; directly mentor 12+ graduate and undergraduate researchers; co-organize STAR-AI Symposium.

Software Engineer Intern — *Mar to Jul 2020*
Cisco Systems, Cyber Vision / IoT Security — *Villeurbanne, France*

- RNN-based anomaly detection for multi-system industrial IoT signals; modeled temporal dependencies and multi-factor state interactions across interconnected machines.

TECHNICAL SKILLS

Languages: Python, C#, C++, SQL **Methods:** Experimental Design, Ablation Studies, Beam Search
AI/ML: PyTorch, TensorFlow, Hugging Face | LLMs, CNNs, GANs, Diffusion Models | Computer Vision
Cloud: Azure (IoT Hub, Function App, Blob Storage, ML), MS Fabric | FHIR, OMOP CDM v5.4 | Docker

Publications: 30+ (9 first/co-first) at ACL Findings, NeurIPS/CVPR/ICLR workshops, IEEE BHI, IEEE ICIR (**Best Paper**), IEEE Reviews in BME, ACM BCB, *Scientific Reports*. Full list: Google Scholar